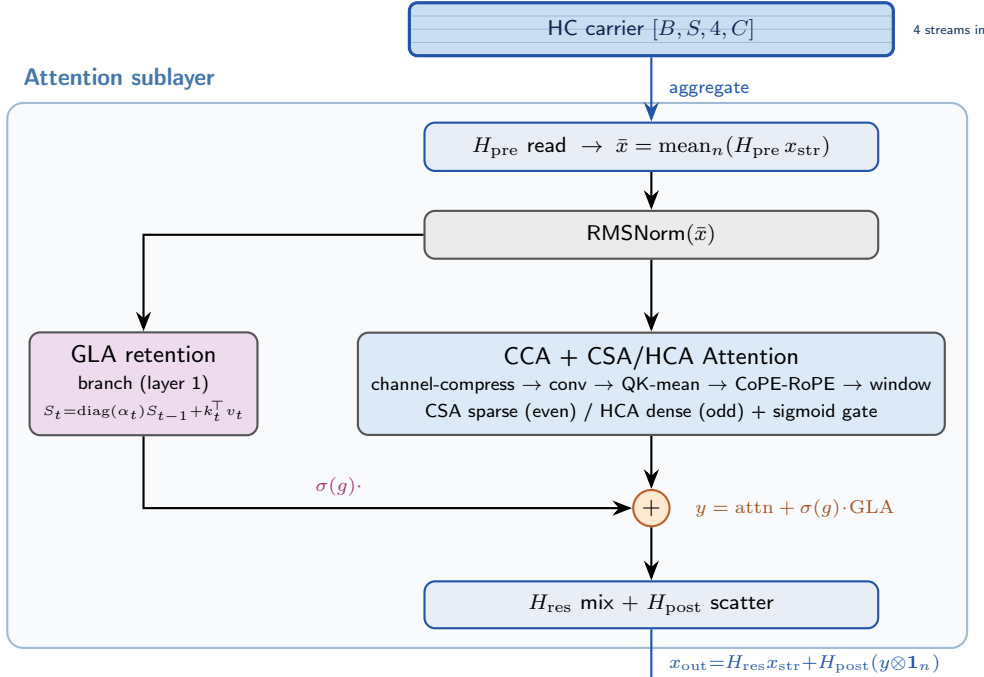


MORPH Block

one block under the Cayley Hyper-Connection carrier

Attention sublayer



HC carrier (Cayley)

The carrier is $n=4$ parallel C -dim streams. Each sublayer:

1. H_{pre} aggregates streams $\rightarrow$ one C -dim $\bar{x}$ that F sees;
2. F (attention or MLP) computes y ;
3. H_{res} (orthogonal, Cayley) mixes streams + H_{post} scatters y .

$H_{\text{pre}}, H_{\text{post}}$ row-stochastic; $H_{\text{res}} = \text{Cayley}(\text{skew}) \in O(n)$. Init $\approx x + F(\text{mean}(x))$ (plain residual).

GLA branch (layer 1 only): runs *parallel* to attention on the same $\bar{x}$; gated sum $y = \text{attn} + \sigma(g) \cdot \text{GLA}$. In core, its state S is carried across loop iterations. $\sigma(g) \approx 0$ at init $\Rightarrow$ off.

MLP sublayer

